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Lecture Notes on Differential Equations and Transform Techniques

Lecture Notes for Spring 2025

The purpose of these lecture notes is to provide a concise summary of the key concepts covered in the course. They are designed to serve as a **quick reference** and should be used in conjunction with the core textbooks listed below. While some sections are adapted from these referenced texts, the notes are not intended to replace the original works, which offer more comprehensive explanations. To gain a deeper understanding, students are encouraged to refer to the textbooks listed.

References:

1. W. E. Boyce, R. C. DiPrima, D. B. Meade, *Elementary Differential Equations and Boundary Value Problems*, 9th edition, Wiley, 1997.

2. Brad G. Osgood, *Lectures on Fourier Transform and Its Applications*, IAS/Park City Mathematics Series / American Mathematical Society (AMS), 2019.

3. Charles Henry Edwards, David E. Penney, *Differential Equations and Boundary Value Problems: Computing and Modeling*, 4th edition, Pearson, 2000.

4. George F. Simmons, *Differential Equations with Applications and Historical Notes*, 2nd edition, CRC Press, 2016.

5. Ruel V. Churchill, *Fourier Series and Boundary Value Problems*, McGraw-Hill, 1941.

6. Gustav Doetsch, *Introduction to the Theory and Application of the Laplace Transformation*, Springer, 2012.

7. Morris Tenenbaum, Harry Pollard, *Ordinary Differential Equations*, Dover Books on Mathematics, Dover Publications, 1985.

EE1060: Differential Equations and Transform Techniques

Lecture Notes - 03

Topics covered :

1. General solution, particular solution and singular Solutions
2. Linear differential equation

- A function $f(x, c_1, \cdots, c_n)$ where $c_1, \cdots, c_n$ are arbitrary constants is called an n-parameter family of solutions of the n-th order differential equation $F(x, y, y', \cdots, y^(n)) = 0$, if for every $c_1, \cdots, c_n$

$$F(x, f, f', \cdots, f^(n)) = 0 \tag{1}$$

- It is not necessary that every differential equation of order n has a solution involving exactly n paramteres. For example, the differential equation

$$\frac{d^2y}{dx^2} + 2a\frac{dy}{dx} + a^2y = 0 \tag{2}$$

has two distinct solutions, each characterized by a different number of parameters. These solutions are $y = ce^{-ax}$ and $y = (c_1 + c_2x)\, e^{-ax}$, , with both functions satisfying the given differential equation, as can be verified by the reader.

- It is also possible for a differential equation to possess multiple solution families, each corresponding to a different number of parameters. For instance [1], the equation

$$(y' - ay)\,(y' - by) = 0 \tag{3}$$

has two distinct solutions, each forming a one-parameter family: $y = c_1e^{ax}$ and $y = c_2e^{bx}$.

- A solution that satisfies a given differential equation and contains no arbitrary constants is termed a particular solution.
- The conditions that facilitate the determination of the arbitrary constants in an n-parameter solution are known as the initial conditions.
- An n-parameter solution that encompasses all particular solutions is referred to as the general solution[1].
- A solution that cannot be derived from an n-parameter solution is called a singular solution.
- Example: The current response of a series LC circuit, given initial conditions for the voltage and current, is governed by the second-order ODE

$$\frac{d^2i}{dt^2} + \frac{1}{LC}i(t) = 0 \tag{4}$$

[1] Recall the arguments covered in the class

The 2-parameter solution to this differential equation, which involves two arbitrary constants, is

$$i(t) = A\cos(\omega t + \phi) \tag{5}$$

where A and ϕ are arbitrary constants and $\omega = \dfrac{1}{\sqrt{LC}}$.

To derive a particular solution, consider the case where the initial current through the inductor is specified as i.e., $i(0) = I_0$, and the initial voltage across the capacitor is given by $v_c(0) = 0$, then a particular solution can be derived. Under these conditions, the constants A and ϕ in the general solution can be determined as follows.

- From the initial condition $i(0) = I_0$, we have

$$i(0) = I_0 A\cos(\omega 0 + \phi) = I_0 \implies A\cos\phi = I_0 \tag{6}$$

- From the initial voltage condition $v_c(0)$, where $v_c(t) = \frac{1}{C}\int i(t)$, we get

$$v_c(0) = \underbrace{\frac{A}{\omega C}\sin(\omega t + \phi)}_{v_c(t)} = 0 \implies \frac{A}{\omega C}\sin\phi = 0 \implies \phi = 0 \tag{7}$$

  Thus, substituting $\phi = 0$ into $A\cos\phi = I_0$, we obtain $A = I_0$

Therefore, the particular solution corresponding to the specified initial conditions is $i(t) = I_0\cos\omega t$.

- A differential equation is linear if it is a linear function of the unknown function and its derivatives. Any linear differential equation can be expressed as

$$a_n(x)y^{(n)} + a_{n-1}(x)y^{(n-1)} + \cdots + a_1(x)y^{(1)} + a_0(x)y = g(x) \tag{8}$$

where $a_i(x)(i = 0, \cdots, n)$ and $g(x)$ are functions of the independent variable x, but do not depend on the dependent variable y or its derivatives.

- Example: Determine whether the following differential equations are linear and identify their order.
  1. $\dfrac{d^2y}{dt^2} = -g - \left(\frac{k}{m}\right)\dfrac{dy}{dt}$ (Linear and second-order; often referred to as a second-order linear ODE)
  2. $L\dfrac{d^2i}{dt^2} + R\dfrac{di}{dt} + \dfrac{1}{C}i(t) = \dfrac{dv(t)}{dt}$ (Second-order linear ODE)
  3. $(1-x^2)\frac{d^2y}{dx^2} + y\frac{dy}{dx} + l(l+1)y = 0$ (Second-order nonlinear ODE)
  4. $\nabla^2\phi(\mathbf{x}) = g(\mathbf{x})$ (Second-order linear PDE) Poisson's Equation

References

[1] M. Tenenbaum and H. Pollard, *Ordinary Differential Equations*, reprint of the 1963 original ed. Dover Publications, 1985, dover Books on Mathematics.